

Model Question Paper Tool Design

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence.

(9x1=9 Marks)

		<i>Module outcome Cognitive level</i>	
1	List out any two examples for Rigid link	M1.01	R
2	List any two forms of thread	M1.02	R
3	What is the function of knockouts?	M2.01	U
4	How is Stopper Important?	M2.01	U
5	List out two examples for Thermoplastics	M3.01	R
6	List types of Compression moulds	M3.01	R
7	What is Melt Flow Index	M3.01	R
8	Give the function of Fixture	M4.01	R
9	What is the function of Bushes in a Jig	M4.01	R

PART B

II. Answer any 8 questions from the following.

(8x3=24 Marks)

		<i>Module outcome Cognitive level</i>	
1	Compare piercing and blanking operation	M1.03	U
2	Differentiate between Kinematic chain and Kinematic Pair	M1.01	U
3	Calculate the blank diameter to make a steel shell having 80 mm outer diameter and 3 cm height with 2 mm thick sheet metal	M2.01	A
4	List the components of die set	M2.01	R
5	Compare thermoplast & thermoset plastic	M3.01	U
6	Explain the importance of venting operation	M3.03	U
7	Explain pot type transfer molding	M3.04	U
8	List any two advantages and one limitations of injection moulding	M3.03	R
9	Give any three advantages of jigs and fixture	M4.01	U

10	List different types of Clamps	M4.02	R
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PART C

Answer all the questions from the following.

(6x7=42 Marks)

Module outcome Cognitive level

III	Explain following sheet metal cutting operation with diagram i. Coining ii. Lancing iii. Planishing	M1.03	U
	OR		
IV	Compare progressive tool with combination tool with rough sketches	M1.04	U
V	Write short note on the following terms a) Kinematic link b) Kinematic pair c) Kinematic chain d) Mechanism	M1.01	U
	OR		
VI	With a neat sketch explain any 4 parameters of a Screw Thread	M1.02	U
VII	Draw a compound tool with direct knock out	M2.02	U
	OR		
VIII	Determine the cutting force for blanking a rectangular piece of size 50 mm x 30 mm from MS sheet 3 mm thickness. Shear strength of material is 320 kg/cm ²	M2.02	A
IX	Compare and explain V bending and Edge Bending with diagram	M2.03	U
	OR		
X	Calculate the bending force for 45° bend in aluminium (shear strength-448 N/mm ²), 0.25 cm thick and 1.12 m long with a die opening 8 times the metal thickness. The bend is to be made with V bending methods	M2.04	A
XI	Illustrate cold chamber die casting	M3.04	U
	OR		
XII	Explain with a neat sketch the working of screw type injection moulding	M3.03	U
XIII	Explain the 3-2-1 principle in jigs and fixture	M4.01	U
	OR		

XIV	Compare the use of jigs with fixtures	M4.01	U
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PART A

I. Answer all questions in one word or one sentence.

(9x1=9 Marks)

Module outcome Cognitive level

1	Define Mechanism	M1.01	R
2	List any 2 Screw thread forms	M1.02	U
3	What is the function of stripper plate in a press tool	M2.01	U
4	Define Strip Layout	M2.02	R
5	How is blank holder pressure significant?	M2.04	U
6	Write any two important properties of plastics	M3.01	R
7	How do shrinkage affect injection molding process?	M3.02	U
8	What are the requirements of a good clamping system	M4.03	R
9	Name any two drilling jigs	M4.04	R

PART-B

II. Answer any 8 questions from the following.

(8x3=24 Marks)

Module outcome Cognitive level

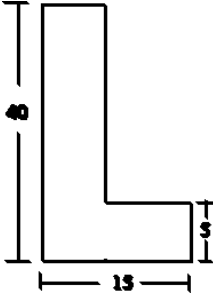
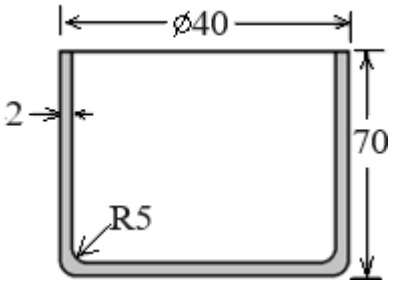
1	Explain Curling operation with a neat figure	M1.03	U
2	Determine the punch and die dimension for blanking a round disk of diameter 0.2 cm from MS sheet 2 mm. Shear strength of material is 278 N/mm ²	M2.01	A
3	Enumerate the steps involved in design of blanking die	M2.01	U
4	List the type of pilots and methods of piloting operation used in press tool	M2.01	R
5	Explain the phenomenon of spring back in bending operation	M2.03	U
6	Explain the need of Gating	M3.04	U
7	Compare Hot Chamber with Cold Chamber Die Casting	M3.04	U
8	List the types of ejectors used in Injection Moulding	M3.01	R
9	List any 3 clamping devices	M4.01	R
10	Write about diamond pin locator with diagram	M4.02	U

PART C

Answer all the questions from the following.

(6x7=42 Marks)

Module outcome Cognitive level

III	Write the equation for calculating degree of freedom of a mechanism and Calculate the degree of freedom for a four- bar chain	M1.01	U
	OR		
IV	Explain combination die set with a neat sketch	M1.04	U
V	Explain any 3 types of thread and their application with neat sketches	M1.02	U
	OR		
VI	Draw and Explain Compound die set	M1.04	U
VII	Illustrate fine blanking with a neat sketch	M2.01	U
	OR		
VIII	An L shaped steel component rectangular piece as in figure is to be made from 3 mm thick sheet. Sketch the scrap strip layout and find out the area utilization. 	M2.02	A
IX	Explain clearance requirement in blanking and piercing operation with a neat diagram	M2.01	U
	OR		
X	Calculate the blank diameter for making a cup as shown in figure 	M2.04	A
XI	Explain with a neat sketch the working of single cavity injection mould	M3.03	U
	OR		
XII	Illustrate and explain about compression mould for component showing vertical flash	M3.04	U
XIII	Explain the components of drilling jig with a neat sketch	M4.01	U

	OR		
XIV	Describe about milling fixture with a neat sketch	M4.04	U